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TOPIC1: Programming Assignments

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TOPIC2: JAVA Interview Question

1. What is a constant in Java and how can you declare a constant variable?
2. What is the purpose of the `final` keyword in Java constants?
3. What are bitwise operators in Java and where are they used?
4. Explain the difference between bitwise AND (`&`) and logical AND (`&&`) operators.
5. What is the purpose of the `main()` method in Java?
6. Why is the `main()` method declared as `public static void main(String[] args)`?
7. Can a Java program run without the `main()` method? Explain.
8. What are command line arguments in Java and how are they passed to a program?
9. How do you access command line arguments inside the `main()` method?
10. What is the `Scanner` class in Java and why is it used?

TOPIC3: JAVA MCQ Question

1. What will be the output of the following code?

```
final int x = 10;  
x = 20;  
System.out.println(x);
```

- A. 10
- B. 20
- C. Compilation Error
- D. Runtime Error

Answer: C. Compilation Error

Reason: `final` variable value cannot be changed.

2. What will this code print?

```
int a = 5, b = 3;  
System.out.println(a & b);
```

- A. 7
- B. 1
- C. 8
- D. 0

Answer: B. 1

Reason: Bitwise AND of 5 (0101) and 3 (0011) = 0001

3. What will be the output?

```
int a = 5, b = 3;  
System.out.println(a && b);
```

- A. 1
- B. 0
- C. Compilation Error
- D. true

Answer: C. Compilation Error

Reason: && works only with boolean values.

4. What is the correct way to declare a constant in Java?

- A. int final x = 10;
- B. final int x = 10;
- C. constant int x = 10;
- D. static int x = final 10;

Answer: B. final int x = 10;

5. What will be printed?

```
public class Test {  
    public static void main(String[] args) {  
        System.out.println(args.length);  
    }  
}
```

Run command: java Test A B C

- A. 0
- B. 1
- C. 3
- D. Error

Answer: C. 3

6. What is the output?

```
int x = 10;  
System.out.println(~x);
```

- A. 10
- B. -11
- C. -10
- D. 11

Answer: B. -11

Reason: Bitwise NOT flips all bits.

7. Which of the following will read input from the user?

- A. System.out
- B. Scanner
- C. String
- D. main

Answer: B. Scanner

8. What will happen if a Java program has no main method?

- A. It runs normally
- B. It throws runtime error
- C. Compilation succeeds but execution fails
- D. JVM ignores program

Answer: C. Compilation succeeds but execution fails

9. What is the output?

```
int a = 10;  
int b = 20;  
System.out.println(a | b);
```

- A. 30
- B. 20
- C. 10
- D. 0

Answer: A. 30

Reason: Bitwise OR of 10 (1010) and 20 (10100) = 11110 (30)

10. What will be the output?

```
import java.util.Scanner;

class Test {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int x = sc.nextInt();
        System.out.println(x + 5);
    }
}
```

Input: 10

- A. 5
- B. 10
- C. 15
- D. Error

Answer: C. 15

TOPIC4: APTITUDE TOPIC

1. Divisibility Problem

Find the smallest number which when divided by 5, 6, and 8 leaves remainder 3 in each case.

2. Remainder Concept

A number divided by 7 leaves remainder 4. What will be the remainder when twice the number is divided by 7?

3. LCM & HCF

The LCM of two numbers is 120 and their HCF is 6. If one number is 24, find the other number.

4. Unit Digit Problem

Find the unit digit of $7^{1037^{103}}7^{103}$.

5. Square Root Concept

Find the least number which must be multiplied with 180 to make it a perfect square.

6. Number Formation

How many 3-digit numbers can be formed using digits 1, 2, 3, 4, 5 without repetition?

7. Prime Factorization

Find the number of trailing zeros in $100!100!100!$.

8. Rational & Irrational Numbers

Determine whether $50\sqrt{50}50$ is rational or irrational. Justify your answer.

9. Divisibility Rule

A number is divisible by 11 if the difference between the sum of digits at odd places and even places is divisible by what number?

10. Average Based Number System

The average of five consecutive even numbers is 24. Find the largest number.